

Arabic & Bilingual Voice — Feasibility for a 3-Day Clinical Copilot

NOTE

State of the art, September 2026. Every number below carries a source and a date. Where a

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number is vendor-self-reported or unverifiable I say so explicitly, and there is a

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[dedicated section](#) listing everything I could not confirm.

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Summary — read this page only

Verdict: GO. A bilingual Arabic/English voice assistant that handles intra-sentence code-switching is buildable in ~3 part-time days in September 2026. It would *not* have been in early 2025. The thing that changed is that code-switching stopped being a research

problem and became a purchasing decision: as of March 2026 there is a commercial, production, real-time **Arabic–English bilingual ASR model — with a medical variant trained on drug names, dosages and ICD-10-CM codes** — which is almost comically on-target for a doctor-facing clinical copilot.

The single biggest risk is the output leg, not the input leg. STT for code-switching is solved by picking the right vendor. TTS is not: ElevenLabs' own docs state that automatic language detection is *"unreliable for short text or codemixed input"* and that production systems must pass explicit [AR] / [EN] tags inline. Your LLM emits untagged mixed text. Something has to tag it before it reaches TTS, or English drug names get read with Arabic phonology (or vice versa) — which in a clinical demo is the exact failure the demo exists to disprove.

Recommended stack for Arabic specifically

STAGE	PICK	WHY
STT (primary)	Speechmatics <code>ar_en</code> bilingual pack, realtime	The only vendor shipping a <i>single</i> bilingual Arabic+English model. Claims 6.3% WER on mixed speech vs Google 9.7%. Gulf + Egyptian + Levantine. Free tier: 480 min/mo, 240 realtime min. First-class Pipecat and LiveKit plugins.
STT (fallback / A-B)	ElevenLabs Scribe v2 Realtime (~150 ms)	Won the only <i>independent</i> commercial code-switching benchmark: 13.2% mean WER vs 38.6% (OpenAI), 39.4% (Google Chirp 3), 43.6% (Azure). Gap widens to 41.5 points on the hardest code-switch quartile.
LLM	GPT-5 / GPT-o3 class, or Claude Sonnet 4	Best published Arabic <i>medical</i> accuracy: 0.765 / 0.764 / 0.694 on MedAraBench (24,883 Arabic MCQs, Feb 2026). Arabic-specialist models are far behind: Fanar 0.498, ALLaM-7B 0.447.
TTS	ElevenLabs Flash v2.5 (Arabic supported, ~75 ms model latency, 137 ms measured median TTFT)	Fastest Arabic-capable option with real voice quality. Requires an explicit language-tagging step.
Do NOT use	Deepgram Nova-3 for code-switching	Deepgram supports 17 Arabic locales, but Arabic is not in Nova-3's multilingual/code-switching mode (that mode is EN/ES/FR/DE/HI/RU/PT/JA/IT/NL only). It is a monolingual Arabic model.

Six facts that should shape the build

1. **Dialect, not Arabic, is the axis of difficulty.** On the same models: MSA 10.19 WER, Levantine 23.53, Egyptian 34.8, **Gulf/Khaleeji 68.39**, Moroccan 81.77 ([Arab Voices, arXiv 2601.13319, 19 Jan 2026](#)). "Arabic works" is meaningless; Gulf is ~6.7× harder than MSA. Demo in Egyptian if you have a choice.
2. **WER is the wrong metric for code-switching and overstates the problem by ~3×.** When a system writes an embedded English word in Arabic transliteration instead of Latin script, WER scores it a 100% substitution error even though it is semantically perfect. BERTScore rankings agree exactly (Kendall's $\tau = 1.000$ on both Arabic pairs) but the magnitudes are far smaller ([arXiv 2605.19069, May 2026](#)). **Consequence for your demo: don't display WER. Show the transcript.**
3. **Script inconsistency is the real failure mode, not error rate.** The same paper's hypothesis for why ElevenLabs' BERTScore *rises* on the hardest quartile (0.929 → 0.938) is that it "consistently picks a single script convention and applies it coherently," while others produce inconsistent output. Your UI has to survive both conventions.
4. **Frontier LLMs beat Arabic-specialist LLMs on medicine, decisively.** MedAraBench and MedArabiQ both find this. MedArabiQ's conclusion: *"task specificity may be more critical than language specificity for Arabic medical tasks."* Don't reach for ALLaM or Jais.
5. **Arabic medical reasoning does lag.** Best model is 0.765 on MedAraBench, which the authors state "does not match expert-level performance." GPT-4's MedArabiQ accuracy collapsed 66.3% → 35.7% under injected cognitive bias. Ship a clinical disclaimer.
6. **`dir="auto"` will break your live captions.** It uses the Unicode *first-strong* heuristic, not majority script. A caption line that begins "patient عندہ chest pain" locks the entire line LTR; the next line beginning in Arabic flips RTL. During streaming the direction can flip mid-token as the first strong character arrives.

Minimum viable demo that still lands

A single live caption pane, Egyptian Arabic in, code-switched transcript rendered with correct bidi, and a **side-by-side toggle: Speechmatics `ar_en` vs Whisper-large-v3**. Speak one sentence — "عندي patient عندہ chest pain ومحتاج aspirin" — and let the audience watch Whisper mangle it while the bilingual model keeps patient , chest pain and aspirin

intact in Latin script. That contrast *is* the demo. Everything else (TTS reply, memory, HUD) is garnish. Budget day 3 entirely for the TTS language-tagging step, because that is where the time will actually go.

1 Arabic STT / ASR quality, September 2026

1.1 The dialect cliff

The most useful single source is **Arab Voices: Mapping Standard and Dialectal Arabic Speech Technology** ([arXiv 2601.13319, 19 Jan 2026](#)), which aggregates 31 datasets across 14 dialects and evaluates 14 open models (Whisper-large-v3, MMS, SeamlessM4T v2, Omnilingual/OmniASR, Qwen3). Best-system WER per variety:

VARIETY	CODE	BEST WER
Modern Standard Arabic	arb	10.19
Levantine	apc	23.53
Egyptian	arz	34.80
Gulf / Khaleeji	afb	68.39
Tunisian	aeb	74.31
Algerian	arq	78.01
Moroccan	ary	81.77
Hassaniyya	mey	88.45

 Gulf Arabic at 68 WER on open models is the finding that should most change your plans. If the take-home is demoed with a Khaleeji speaker on an open model, it fails. The commercial bilingual models (§2) are the only credible answer for Gulf.

Note: this paper evaluates **open models only — no commercial APIs**. Datasets used include Common Voice, FLEURS, MGB-2, MGB-3, MGB-5, MASC, QASR, GALE, Casablanca and SADA; the code-mixed sets it catalogues are **ArzEn, Mixat, SCC and ZAEBUC-Spoken**.

1.2 Open-weights leaderboard (multi-dialect, zero-shot)

The [Open Universal Arabic ASR Leaderboard](#) (ELM Research Center; methodology published at [Interspeech 2025](#)) evaluates zero-shot multi-dialect generalisation over six test sets spanning MSA, Egyptian, Gulf, Levantine and Maghrebi.

MODEL	AVG WER	COMMONVOICE	MASC - C	MASC - N	MGB - 2	SADA	CASABLANCA
Audar-ASR-V1-Turbo	24.78	8.60	19.60	28.35	11.13	29.41	51.58
Cohere Transcribe Arabic	25.87	5.82	15.54	27.07	—	37.47	49.71
Munsit-1 (CNTXT AI)	26.68	—	—	—	—	—	—
OmniASR-LLM-7B	28.32	9.75	19.69	29.29	—	41.61	56.46
Qwen3-ASR-0.6B	42.18	—	—	—	—	—	—
Whisper Large v3	36.86	17.83	24.66	34.63	—	55.96	71.81

Reading this table:

- **Whisper large-v3 is ~11 WER points behind the Arabic-specialist state of the art**, and the gap is worst exactly where it matters — Casablanca (multidialectal) 71.81 vs 49.71.
- **Casablanca is brutal for everyone**. Even the leader is at ~50 WER. Casablanca ([arXiv 2410.04527](#)) is the honest multidialectal test; Common Voice AR is the flattering one (5.8–8.6 WER). Beware anyone quoting only Common Voice.
- **Audar-ASR-V1** explicitly ships Arabic-English code-switching via a `<tag:ar+en>`

control token, trained on 300k+ hours; weights on Hugging Face, API invite-only. Its internal Gulf-Emirati set scores 30.02 WER — far better than the 68 above, but it is a *vendor-internal* set, so not comparable.

- **Cohere Transcribe Arabic** is Apache 2.0, free on Cohere's API with rate limits, RTFx 525 (vs Whisper's 146), and claims native-speaker preference over Whisper in 95.8% of tests on accuracy, dialect faithfulness and code-switching robustness. It preserves the speaker's dialect rather than normalising to MSA.

1.3 Commercial providers — what each actually offers for Arabic

PROVIDER	ARABIC STATUS	DIALECTS	REALTIME	PUBLISHED ARABIC WER
Speechmatics	ar + ar_en bilingual pack ; medical variant	Gulf, Egyptian, Levantine, MSA	Yes, sub- second, same model as batch	4.5% Arabic-only; 6.3% code-switched (vendor)
ElevenLabs Scribe v2 / v2 Realtime	Supported; docs tier Arabic as " Good: >10% to ≤20% WER "	Not broken out	Yes, ~150 ms	Tier only, no point estimate
Deepgram Nova-3	17 Arabic locales (ar , ar-EG , ar-SA , ar-AE , ar-IQ , ar-MA ...)	Gulf, Levantine, Egyptian/Nile, Maghrebi, Mesopotamian	Yes (streaming + batch)	None published; blog claims "~40% lower WER" unqualified
Soniox v5	60+ languages incl. Arabic, mid-sentence switching	Not broken out	Yes, sub- 200 ms	None published
OpenAI (Whisper / gpt-4o- transcribe)	Yes	Poor on dialect	Realtime API yes	36.86 avg (Whisper, leaderboard)
Google Chirp 3	Yes, auto language inference	Weak on Gulf (see §2)	Yes	None published

PROVIDER	ARABIC STATUS	DIALECTS	REALTIME	PUBLISHED ARABIC WER
Azure AI Speech (CLID)	Yes, segment-level LID	Multiple locales	Yes	None published
AssemblyAI Universal 3.5 Pro	Yes	Not broken out	Yes	40.9% mixed error (per Speechmatics)
Munsit / CNTXT AI	Arabic-only specialist, 25+ dialects	All major	Yes, "sub-300 ms" (vendor)	26.68 avg (leaderboard)
Audar AI	Arabic-first, open weights	Multi	Not stated	24.78 avg (leaderboard)
Cohere Transcribe Arabic	Open weights, Apache 2.0	Multi	Not stated	25.87 avg (leaderboard)

Sources: [Speechmatics languages docs](#), [ElevenLabs STT docs](#), [Deepgram models & languages](#), [Soniox Arabic](#), [Deepgram Nova-3 Arabic blog](#). All retrieved 18 Sep 2026.

2 Code-switching (Arabic ↔ English, intra-sentence) — the make-or-break section

2.1 The one independent benchmark

Benchmarking Commercial ASR Systems on Code-Switching Speech: Arabic, Persian, and German ([arXiv 2605.19069](#), v1 18 May 2026, v3 22 May 2026). Four language pairs — Egyptian Arabic–English, Saudi Arabic (Najdi/Hijazi)–English, Persian–English, German–English — 300 samples each, natural conversational recordings by native speakers, difficulty-stratified by a computed `H_Score` .

Overall results (Table 4):

SYSTEM	MEAN WER	MEAN BERTSCORE	CS PAIRS COVERED
ElevenLabs Scribe v2	13.2%	0.936	4
OpenAI gpt-4o-transcribe	38.6%	0.856	4
Google Chirp 3	39.4%	0.862	4
Azure AI Speech (CLID)	43.6%	0.839	4
Deepgram Nova-3	5.0% †	0.959 †	1 (German only)

† The paper explicitly excludes Deepgram from the ranking: *"Deepgram Nova-3 does not list Arabic or Persian in its documented CS language support. For these pairs we suppress WER and BERTScore and exclude the system from aggregate rankings."* Its 5.0% is German-only and is **not** an Arabic result. Do not quote it as one.

By difficulty quartile (Table 5) — this is the table that matters:

QUARTILE	ELEVENLABS	GOOGLE CHIRP 3	OPENAI	AZURE
Q1 (easiest)	2.0%	4.4%	9.7%	17.1%
Q2	13.9%	30.8%	46.2%	46.4%
Q3	15.0%	54.4%	48.7%	54.2%
Q4 (hardest)	20.0%	61.5%	45.2%	52.2%

⚠ The 2.4-point gap between ElevenLabs and Google at Q1 becomes a **41.5-point gap at Q4**. Q4 is dominated by Arabic and Persian utterances with dense switching and morphological blending — i.e. exactly how your doctors speak. Aggregate WER numbers hide this completely. Never choose an Arabic code-switching ASR on an aggregate number.

On Gulf specifically: *"Google Chirp 3 performs worse on Saudi Arabic than on Egyptian Arabic, suggesting that Gulf dialectal phonology remains less robustly covered than more widely represented Arabic varieties."*

2.2 What actually happens to the English words

Three distinct behaviours, all documented:

1. **Kept in Latin script** — the good case. ElevenLabs is reported to pick one script convention and apply it coherently.
2. **Transliterated into Arabic script** — semantically correct, *scored as 100% wrong by WER*. This is the source of the $\sim 3\times$ inflation. It is also a genuine UI problem for you: `أسبرين` and `aspirin` must both map to the same drug downstream.
3. **Failure modes at switch boundaries** — the paper's architectural analysis notes that segment-level LID systems (Azure CLID) are *"structurally limited to producing one script per recognition segment"*, so a mid-segment switch cannot be represented at all.

Whisper specifically has a well-documented additional pathology: it will return **Arabic speech transliterated into Latin script** even with `language="ar"` set (e.g. "marhaban" instead of "مرحبا") — reported repeatedly on the [OpenAI developer forum](#). Fine-tunes exist specifically to patch this (e.g. [Arabic Whisper CodeSwitching Edition](#)), which is itself evidence base Whisper is inadequate here.

2.3 Named code-switching benchmarks

- **ArzEn** (Hamed, Vu & Abdennadher 2020) — Egyptian Arabic–English, 6,213 sentences, **66.9% code-mixed**, 30.0% monolingual Arabic, 3.1% monolingual English. Extended by **ArzEn-ST** ([arXiv 2211.12000](#)) into a three-way speech-translation corpus with ASR/MT/ST baselines, and used by **ArzEn-LLM** ([arXiv 2406.18120](#)).
- **Mixat** — Emirati Arabic–English code-mixed; catalogued in Arab Voices.
- **SCC, ZAEBUC-Spoken** — additional code-mixed sets in the Arab Voices catalogue.
- **SwitchLingua** (Xie et al. 2026) — 12-language synthesized+recorded CS benchmark.
- **SAGE** ([arXiv 2506.22143](#)) — spliced-audio data augmentation for low-resource Arabic-English CS ASR.
- The 2605.19069 benchmark set itself is published on Hugging Face.

2.4 The vendor that built for exactly this

[Speechmatics](#) (6 Mar 2026) shipped a single bilingual Arabic–English model rather than two monolingual ones. Claims: **6.3% WER on mixed speech** (vs Google 9.7%, "35% fewer errors") and **4.5% on Arabic-only** (vs Google 5.9%, Whisper 6.2%). Gulf + Egyptian + Levantine. Sub-second latency, same model realtime and batch, cloud / on-prem / on-device.

Four days later ([10 Mar 2026](#)) they announced an **Arabic–English medical** variant: trained on twice the vocabulary of their English medical model, covering **ICD-10-CM codes, drug names, dosages and clinical shorthand** "regardless of which language carries them", built from real clinical audio with dialect variation. Their own framing of the problem ([5 Mar 2026](#)) is almost verbatim your brief: *"A doctor names a drug in English and finishes the sentence in Arabic."*

Their newer general model **Melia 1** (launched June 2026) was benchmarked [1 Sep 2026](#) at **15.1% mixed error rate on Arabic–English with no language hints**, vs Soniox v5 33.2%, AssemblyAI Universal 3.5 Pro 40.9%, Amazon 45.6%. Vendor-run, unnamed test set — treat as directional, not as ground truth.

Practical config. The [Speechmatics languages docs](#) list an **"Arabic & English bilingual" pack, language code `ar_en`**, described as *"ideal when transcribing Arabic and English in the same media file or stream."* Note a documentation conflict: the [Pipecat Speechmatics service docs](#) document bilingual modes via `language + domain` (e.g. `Language.ES + domain="bilingual-en"`, plus packs `cmn_en`, `en_ms`, `en_ta`) and **do not list `ar_en`**. Resolve this against the live API on day 1 — try `ar_en` as a language code first, then `language="ar", domain="bilingual-en"`. Plugins exist for both [LiveKit](#) (`livekit-plugins-speechmatics`) and Pipecat.

3 Arabic TTS, September 2026

3.1 Latency

Measured TTFT, [SILMA benchmark, 15 Sep 2026](#):

MODEL	MIN TTFT	MEDIAN TTFT	COST/MIN
Cartesia Sonic 3.5	102 ms	120 ms	—
ElevenLabs Flash v2.5	131 ms	137 ms	\$0.050
Deepgram Aura-2	251 ms	256 ms	\$0.027
SILMA TTS v2	279 ms	280 ms	\$0.025
ElevenLabs v3	677 ms	730 ms	\$0.100

⚠ This benchmark does **not** break results out by language. These are global TTFT figures, not Arabic-specific. Arabic TTFT is unverified for every provider.

ElevenLabs' own docs claim ~75 ms model latency for Flash v2.5; the 137 ms measured figure is the honest one to plan against.

3.2 Dialect coverage — this is where providers diverge sharply

PROVIDER	ARABIC DIALECT COVERAGE	SOURCE
Azure Neural TTS	~20 locales: ar-SA , ar-EG , ar-AE , ar-LB , ar-OM , ar-IQ , ar-MA ... named voices each (Salma/ar-EG, Zariyah/ar-SA)	MS Learn
ElevenLabs Multilingual v2	Arabic listed as "Arabic (Saudi Arabia, UAE)"; Egyptian/Gulf/Moroccan voices exist in the voice library	ElevenLabs models
ElevenLabs Flash v2.5	Arabic among 32 languages	same
ElevenLabs v3	Arabic among 70+ languages	same
Google Cloud (Chirp 3 HD)	ar-XA — MSA only. No Gulf, Levantine, Egyptian or Maghrebi voice variants documented	Google TTS voice list
Amazon Polly	MSA + Gulf (ar-AE)	Munsit TTS roundup, 30 Jul 2026
Cartesia Sonic 3 / 3.6	Arabic among 42 / 44 languages; 3.6 lists "several Arabic locales"	Cartesia docs

PROVIDER	ARABIC DIALECT COVERAGE	SOURCE
Munsit / Faseeh TTS	Gulf, MSA, Levantine, Egyptian, Maghrebi + dedicated Tashkīl (diacritization) API endpoint	Munsit roundup
Soniox TTS	Arabic voices, "starts speaking in under a second", explicitly handles "alphanumerics, names, borrowed words, language switching"	Soniox Arabic

3.3 Diacritization — the structural Arabic TTS problem

Arabic script omits short vowels. `كتب` is *kataba* ("he wrote"), *kutub* ("books"), or *kutiba* ("it was written") — same letters, three pronunciations, three meanings. TTS must restore diacritics (tashkīl) before phonemisation, and this is a genuine NLP task, not a lookup.

- Microsoft reports that ****improving diacritic prediction alone produced a 78% reduction in pronunciation errors**** for `ar-SA` and `ar-EG` voices ([Azure AI blog](#)).
- 2026 research (KSAA-2026 shared task at OSACT7/LREC 2026) finds ****text-only Seq2Seq diacritizers outperform off-the-shelf multimodal/ASR-based ones**** ([Fine-Tashkeel, ACL Anthology](#); [Thaka, arXiv 2605.25928](#)).
- Practical escape hatches: Munsit exposes diacritization as its own API so you can fix pronunciation deterministically instead of regenerating audio; Azure and Polly accept SSML with IPA phonemes; ElevenLabs' route is phonetic respelling.

3.4 Embedded English inside an Arabic sentence — the actual risk

ElevenLabs' documentation is explicit: language detection is **"automatic but unreliable for short text or codemixed input — always pass explicit language tags in the text parameter (e.g. [EN] Hello [ES] Hola) for production systems to avoid mispronunciation"**, and further warns to *"avoid mixing right-to-left scripts (Arabic, Hebrew) without explicit markup."*

 **This is the single biggest implementation risk in the project.** Your LLM will emit `"مرتين يومياً aspirin المريض محتاج"` as plain text. Handed to TTS untagged, the English token is a coin flip. You need a tagging step between LLM and TTS: either (a)

instruct the LLM to emit `[AR]` / `[EN]` span tags directly in its output, or (b) run a cheap regex on Unicode script ranges (Arabic block `U+0600–U+06FF` vs Latin) to segment and tag. Option (b) is ~20 lines, deterministic, and is what I would ship in 3 days.

Independent corroboration that this is a real production problem: TTS providers "mispronounce names, addresses, medical terms, product names, acronyms" and *"the mispronunciation isn't even consistent — the model will pronounce the same word differently across calls"* ([Rime, TTS best practices 2026](#)). Hence the existence of pronunciation-dictionary features ([Telnyx](#)) — worth wiring a small dictionary of your demo's drug names if time permits.

4 Realtime speech-to-speech APIs and Arabic

OpenAI Realtime

- Arabic **is** supported. `gpt-4o-realtime` documents 70+ input languages; OpenAI states it only lists languages that came in **under 50% WER**, and *"the model returns results for languages not listed but the quality will be low."* Arabic makes the list, but that is a low bar — a 50% WER threshold tells you nothing about Egyptian vs Gulf.
- `gpt-realtime-translate` supports 70+ input / 13 output languages, with Arabic named among the better-performing output languages ([OpenAI cookbook](#)).
- **No dialect breakdown is published anywhere.** There is no `ar-EG` vs `ar-SA` story.
- Known pathology: with `input_audio_transcription` enabled, ****Arabic speech sometimes comes back transcribed as English****. Workaround is to set the `language` parameter explicitly on `input_audio_transcription` ([OpenAI forum, 10 Feb 2025](#)). Old report, but the underlying language-lock behaviour is the same class of bug the code-switching benchmark documents.

Gemini Live

- Arabic **is** in the documented list — **ar** , **one of 99 supported languages** ([Live API capabilities](#), retrieved 18 Sep 2026). **No regional variants** (ar-EG , ar-XA) are documented.
- Native-audio models *"can switch between languages naturally during conversation"* — the closest thing to a documented code-switching claim from either realtime vendor, though it is about turn-level switching, not intra-sentence.
- **Stability is a live concern, independent of language.** Through 2026 the Google AI developer forum carries reports of: audio regression after the 9 Mar 2026 update; voice/volume degradation on outputs longer than ~1 min; premature `turnComplete` causing [mid-sentence truncation](#); and, as recently as **5 Sep 2026**, **first-audio latency of 16–26 s** on `gemini-3.1-flash-live-preview` ([forum thread](#)).

⚠ **Recommendation: do not build the Arabic demo on a speech-to-speech API.**

Neither vendor publishes Arabic dialect quality, neither lets you swap the ASR, and a black-box S2S pipeline gives you nothing to *show*. A cascade lets you put the code-switched transcript on screen — which is the whole point. Keep Gemini Live as an optional toggle if you already have one wired from the English build.

5 Arabic LLM quality for medical reasoning

5.1 MedAraBench — the large one

[arXiv 2602.01714](#), 2 Feb 2026. **24,883 Arabic multiple-choice medical questions**, 19 specialties, five difficulty levels (Y1–Y5), digitised from regional medical school exams, quality-checked by clinicians and LLM-as-judge.

MODEL	ACCURACY
GPT-o3	0.765
GPT-5	0.764

MODEL	ACCURACY
Claude Sonnet 4	0.694
GPT-4.1	0.673
Gemini 2.0 Flash	0.654
DeepSeek-Chat-v3	0.620
Qwen-Plus	0.618
Llama-3.3-70B	0.547
Fanar-C-1-8.7B (Qatar)	0.498
ALLaM-7B (Saudi)	0.447
MedGemma-4B	0.390

Two conclusions, both actionable: **frontier general models win by 25+ points over Arabic-specialist models**, and even the winner "does not match expert-level performance."

5.2 MedArabiQ – the diagnostic one

[arXiv 2505.03427](https://arxiv.org/abs/2505.03427). Seven Arabic medical tasks (MCQ, fill-in-blank, patient–doctor Q&A).

Findings that matter for a clinical copilot:

- **No single model dominates.** Gemini 1.5 Pro 57.5% and Claude 3.5 Sonnet 53.5% lead on MCQ; DeepSeek v3 79.7% on fill-in-blank-with-choices; **Jais 85.7% on open-ended patient–doctor Q&A** – an Arabic-specialist model winning the *conversational* task while losing the knowledge tasks. If your copilot is dialogue-shaped, that is worth knowing.
- **Cognitive bias is devastating.** GPT-4 fell from **66.3% → 35.7%** under bias-injected conditions. A copilot that echoes a doctor's stated hypothesis will amplify anchoring.
- **The benchmark is MSA-only**, which the authors flag as a limitation: it "may not reflect real clinical communication using dialects." So the published numbers are an *upper bound* on real dialectal clinical performance.
- Authors' conclusion: *"task specificity may be more critical than language specificity for Arabic medical tasks."*

5.3 Other Arabic medical evaluation

- **AraHealthQA 2025** ([ACL Anthology](#)) —

first shared task on Arabic health QA, two tracks: MentalQA and MedArabiQ, the latter spanning 12 specialties. Winning approaches were prompt-engineering + ensembling over **Gemini 2.5 Flash**, not fine-tuned Arabic models ([arXiv 2509.11365](#)).

- **Arabic Healthcare Understanding & Reasoning** ([arXiv 2508.15797](#)).
- **QIMMA** ([arXiv 2604.03395](#)) argues many Arabic benchmarks are themselves unreliable — worth a caveat line in your write-up.

5.4 Arabic-specialist LLMs, for completeness

- **Falcon-H1-Arabic** (TII, [5 Jan 2026](#)) —

hybrid Mamba-Transformer; 7B scores 71.47% avg on the Open Arabic LLM Leaderboard (beating Fanar-1-9B and ALLaM-7B), 34B scores 75.36% (beating Qwen2.5-72B and Llama-3.3-70B). Strong on 3LM (STEM), ArabCulture, AraDice (dialect).

- **Fanar 2.0** (QCRI/HBKU, [arXiv 2603.16397](#)) —
Fanar-2-27B-Instruct, Apache 2.0, ~Mar 2026.
- **ALLaM** — now under HUMAIN (Saudi).

These are excellent for Arabic *fluency and culture* and poor for *medicine*. For a clinical copilot, use a frontier model and spend your effort on the prompt.

6 RTL / bilingual frontend

6.1 The core bug: `dir="auto"` uses first-strong, not majority

The HTML `dir="auto"` attribute resolves direction from the **first strong directional character** in the element, per the Unicode Bidirectional Algorithm. It is not a vote. [Direction is not alignment \(13 Sep 2026\)](#) puts it precisely: "*Appending Persian after an existing unisolated Hello does not make it switch merely because Persian now dominates.*"

For your live captions this means: **a caption line that begins with an English word renders entirely LTR**, punctuation lands on the wrong side, and the Arabic reads right-aligned in a left-aligned block. The next line, beginning in Arabic, flips. The pane visually stutters.

Worse during streaming: tokens arrive incrementally. If the first token is Arabic the line commits RTL; if the first arriving token happens to be `patient` it commits LTR and the rest of the Arabic reflows under an LTR paragraph. Direction can visibly flip mid-stream.

6.2 Fixes, in order of effort

1. **Compute direction per line from dominant script, not first character.** Count Arabic-block codepoints (`U+0600–U+06FF` , plus `U+0750–U+077F` , `U+FB50–U+FDFF` , `U+FE70–U+FEFF`) vs Latin, set `dir` explicitly. This is the single highest-value fix and is ~15 lines. [`bidirectional`](#) is a zero-dependency library that does exactly this — "resolve text direction by dominant script, not by the first character; fixes mixed Persian/Arabic/Hebrew text that `dir="auto"` renders backwards."
2. **Isolate every foreign-script span with `<bdi>`** . Wrap each Latin run inside an Arabic line: `<p dir="rtl">عندي <bdi dir="ltr">patient</bdi> عنده <bdi dir="ltr">chest pain</bdi></p>` . `<bdi>` is exactly "treat this text in isolation from its surroundings" ([MDN/CodeLucky](#)). Equivalent CSS is `unicode-bidi: isolate` .
3. **Separate the three concerns.** Direction (`dir`), alignment (`text-align`) and isolation (`<bdi>`) are independent. The named "critical mistake" is conflating direction with alignment. You can legitimately want `dir="rtl"` with `text-align: left` in a caption rail.
4. **Freeze direction once committed during a stream.** Decide the line's `dir` on first flush and don't recompute on every token, or accept flicker.
5. **Next.js specifics.** Next does **not** set direction from `lang` — you must set `document.documentElement.dir` yourself, usually from your `i18n` layer ([next-i18next discussion #1738](#)). Use CSS logical properties (`margin-inline-start` , `padding-inline-end` , `inset-inline`) throughout rather than left/right so one `dir` flip restyles the whole app; Tailwind's `ms-* / me-* / ps-* / pe-*` and `rtl: / ltr:` variants cover this. ([LeanCode RTL in React](#), [better-i18n RTL guide](#))
6. **If you want the heavy artillery, [BidiLens](#)**

bundles per-block direction policy, inline isolation, **streaming stability**, and bidi-control auditing with React adapters. Probably overkill for 3 days, but its existence tells you streaming bidi is a known hard problem.

6.3 Security note worth one line in the write-up

Unicode bidi control characters (RLO/LRO, U+202A – U+202E) can visually reorder text without changing its bytes — the "Trojan Source" class of attack. If transcripts are ever rendered into a clinical record, strip bidi control characters from ASR output before storage. BidiLens ships "bidi-control auditing" for this reason.

7 Verdict

7.1 Is it buildable in ~3 part-time days on free/cheap tiers?

Yes — GO. The decisive change is that Arabic-English code-switching is now a line item you buy rather than a model you train. Concretely:

- **Speechmatics free tier:** 480 min/month total, **240 realtime minutes**, 2 concurrent realtime sessions, 3,000 Voice Agent minutes. Pro is \$0.0067/min realtime (~\$0.40/hr). ([Speechmatics pricing](#) via [PulseSignal](#), retrieved 18 Sep 2026 — **verify on the pricing page directly; tier details move.**)
- **Groq Whisper large-v3-turbo free tier:** 20 RPM, 2,000 req/day, 28,800 audio sec/day, \$0.04/hr paid — the cheapest possible *baseline* to A/B against ([Groq docs](#)).
- **Cohere Transcribe Arabic:** free on Cohere's API with rate limits, Apache 2.0 weights.
- **Soniox:** \$0.12/hr realtime STT, \$0.10/hr async, ~\$0.70/hr TTS, no free tier documented.
- **Munsit:** free 10,000 credits (~30 min), then \$8/mo; \$0.12/hr realtime.
- **ElevenLabs:** Scribe realtime from ~\$0.28/hr; TTS Flash v2.5 ~\$0.05/min.

A three-day budget of a few dollars covers the whole build.

7.2 The single biggest risk

The TTS output leg — specifically, rendering an LLM-generated Arabic sentence containing English drug names without mispronouncing them. Ranked reasoning:

1. STT code-switching: **solved by vendor choice**. Low risk.
2. TTS code-switching: **not solved by vendor choice**. ElevenLabs' own docs say automatic detection is unreliable on codemixed input and mandate explicit tags. You must build the tagging step. Unknowns compound: diacritization ambiguity on the Arabic, phoneme choice on the English, and no published Arabic-specific TTFT from anyone.
3. Second-order risk: **240 free realtime minutes disappears fast** when you're re-running the same test sentence 200 times. Record 5–10 canonical WAV files on day 1 and iterate against files, not a live mic.
4. Third: **Gulf dialect**. If your demo audio is Khaleeji rather than Egyptian, every number in §1.1 gets worse. Control this — pick your demo speaker.

7.3 Minimum viable version that still impresses

Strip to the contrast, not the pipeline:

1. **Mic → Speechmatics `ar_en` realtime → caption pane with correct bidi**. That alone is the demo.
2. **A/B toggle to Whisper-large-v3 via Groq** on the same audio. Side by side. The audience sees `chest pain` survive on one side and dissolve into Arabic transliteration on the other. This is the single most persuasive 20 seconds you can build.
3. **One LLM turn** — a clinical summary or a drug-interaction flag — proving the code-switched transcript is machine-usable downstream (e.g. `aspirin` and `أسبرين` normalise to the same RxNorm-style key).
4. **TTS reply with the script-range tagger** (§3.4 option b). Even a 20-line regex tagger demonstrates you understood the problem.
5. **Skip**: speech-to-speech, dialect ID, diacritization models, Arabic-specialist LLMs, any fine-tuning. All are day-5+ work.

7.4 Ranked stack for Arabic

RANK	STACK	WHEN TO PICK IT
1	Speechmatics ar_en realtime → GPT-5/o3 → ElevenLabs Flash v2.5 + script tagger	Default. Only true bilingual model, medical variant exists, free tier sufficient, plugins in both Pipecat and LiveKit.
2	ElevenLabs Scribe v2 Realtime (~150 ms) → GPT-5 → ElevenLabs Flash v2.5	If Speechmatics' ar_en config fights you on day 1. Independently benchmarked best on code-switching; one vendor for STT+TTS; ~150 ms.
3	Soniox v5 realtime (sub-200 ms, \$0.12/hr) → GPT-5 → Soniox TTS	Cheapest coherent single-vendor bilingual option; explicitly claims mid-sentence switching and handles "borrowed words". Weaker third-party validation (33.2% mixed error in Speechmatics' own test).
4	Groq Whisper-v3-turbo → GPT-5 → ElevenLabs	Baseline only. Use as the losing side of the A/B. Not viable as the product.
5	Gemini Live native audio (Arabic ar)	Optional toggle if already wired. No dialect docs, 2026 stability complaints, and it hides the transcript — which is your demo.
—	Deepgram Nova-3 Arabic	Avoid for this use case. 17 Arabic locales but Arabic is excluded from the code-switching multilingual mode. Fine for monolingual Arabic.

8 What I could not verify

Listed explicitly, as requested.

CLAIM	STATUS
Per-language-pair WER for Egyptian and Saudi Arabic in arXiv 2605.19069	NOT VERIFIED. The paper presents these only in Figures 2 and 3 (raster images); the text says "Figures 2 and 3 report the per-language-pair results, making separate result tables unnecessary." I have the overall means (Table 4) and the difficulty quartiles (Table 5) verbatim from the PDF. An earlier automated read of the PDF produced per-pair numbers around "45–50% for OpenAI on Egyptian" — I could not confirm those in the extracted text and have excluded them. Do not cite them.

CLAIM	STATUS
Speechmatics 6.3% / 4.5% / 15.1% figures	Vendor self-reported , test set unnamed, not independently reproduced.
Melia 1 competitor numbers (Soniox 33.2%, AssemblyAI 40.9%, Amazon 45.6%)	Vendor-run comparison , unnamed test set.
Deepgram Nova-3 Arabic WER	No published number exists. The "~40% lower WER" blog claim has no named benchmark and no date on the page.
ElevenLabs Arabic-specific WER	Only a docs <i>tier</i> : "Good, >10% to ≤20% WER". No point estimate, no dialect breakdown.
Munsit "sub-300 ms realtime" and "<150 ms TTS"	Vendor self-reported. No independent measurement found.
Arabic-specific TTS TTFT for any provider	NOT VERIFIED. The SILMA 15 Sep 2026 benchmark gives global TTFT only, not per-language.
Whether the Speechmatics medical Arabic-English model is available self-serve / on the free tier	NOT VERIFIED. Press release says "available now in our portal"; no pricing or tier detail found. Check before relying on it.
Exact Speechmatics bilingual Arabic config parameter	CONFLICTING. Speechmatics docs list pack <code>ar_en</code> ; Pipecat's service docs list bilingual packs <code>cmn_en / en_ms / en_ta</code> + <code>domain="bilingual-en"</code> and do not include <code>ar_en</code> . Resolve against the live API.
Whether Scribe v2 Realtime covers the same 90+ languages as batch Scribe v2	NOT STATED in ElevenLabs docs. Both are described as "90+ languages" but no realtime language list is published.
Arabic quality reports for Gemini Live specifically	NONE FOUND. All 2026 forum complaints are language-agnostic (compression, truncation, latency regressions).
Arabic quality reports for OpenAI Realtime specifically	Only a Feb 2025 forum report of Arabic being returned in English. No 2026 Arabic-specific evaluation found.

CLAIM	STATUS
Whisper large-v3 vs large-v3-turbo Arabic difference	NOT FOUND. No source compares the two on Arabic. Leaderboard figures are for large-v3.
Modern commercial WER on MGB-2 / MGB-3	Only Audar-ASR-V1-Turbo publishes MGB-2 (11.13). No commercial API publishes MGB-2 or MGB-3.
Gulf-dialect WER for any commercial provider	NOT PUBLISHED by anyone. The 68.39 figure is open models only. Audar's 30.02 Gulf-Emirati number is a vendor-internal test set.
Speechmatics free-tier exact limits	Sourced from a third-party pricing aggregator, not the vendor page. Verify directly.
Cartesia Sonic Arabic <i>quality</i>	Language support is documented (42→44 languages incl. Arabic); no Arabic quality or dialect evaluation exists.
Artificial Analysis "Scribe v2 first at 2.3% AA-WER (Feb 2026)"	This is an aggregate multilingual figure surfaced via search summary; I did not verify it on Artificial Analysis directly and it is not an Arabic number.

Sources

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- [ArzEn-ST, arXiv 2211.12000](#) · [ArzEn-LLM, arXiv 2406.18120](#) · [SAGE, arXiv 2506.22143](#)
- [Fine-Tashkeel, KSAA-2026](#) · [Thaka, arXiv 2605.25928](#)
- [Direction is not alignment](#) — 13 Sep 2026 · [bidi-dir](#) · [BidiLens](#) · [HTML bdi](#) · [RTL in React \(LeanCode\)](#) · [next-i18next direction discussion](#)
- [Rime TTS best practices 2026](#) · [Telnyx pronunciation dictionaries](#)

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